The Exact Partially Symmetric Rank of Products of W-States

A proof via moment conditions and weighted Gram matrices

Wenqi Zhu*

Mathematical Institute, Woodstock Road, University of Oxford,

Oxford, UK, OX2 6GG

wenqi.zhu@maths.ox.ac.uk

Ping Nie

David R. Cheriton School of Computer Science, University of Waterloo

ping.nie@uwaterloo.ca

Abstract

We study the minimum number of products of powers of binary linear forms needed to express $\prod_{i=1}^{k} x_i^{d_i-1}y_i$. For $k \geq 2$ and $d_i \geq 3$, we prove that this number is $2^{k-1}(\sum_i d_i - 2k + 2)$, matching the upper bound of Canino, Casarotti and Santarsiero. The lower bound starts from an arbitrary complex decomposition. A change of coordinates converts its coefficients into weighted points satisfying product moment conditions. We construct rational test functions whose evaluation vectors have a block diagonal weighted Gram matrix. Its blocks are nondegenerate at one finite choice of auxiliary parameters, so the number of decomposition terms is at least the number of test functions. The argument allows repeated points and zero coordinates and uses no positivity assumption. We also determine the exact partially symmetric rank of every product of binary monomials $x_i^{a_i}y_i^{b_i}$ with $a_i \geq b_i \geq 1$, with a matching explicit decomposition.

1 Introduction and main result

For $\mathbf{d} = (d_1, \ldots, d_k)$, consider the multihomogeneous polynomial

$$F(x, y) = \prod_{i=1}^{k} x_i^{d_i-1}y_i. \tag{1}$$

The pair (x_i, y_i) represents the ith symmetric tensor block, of order d_i. Thus F represents a product of k generalized W-states, up to a nonzero normalization scalar. In this polynomial description, a rank-one term with respect to the prescribed blocks is a product $\prod_i (a_i x_i + b_i y_i)^{d_i}$.

Definition 1.1 (Partially symmetric rank)**.** The partially symmetric rank $R_{\mathbf{d}}(F)$ is the least integer r for which there are complex coefficients $c_\ell, a_{i\ell}, b_{i\ell}$ satisfying

$$F(x, y) = \sum_{\ell=1}^{r} c_\ell \prod_{i=1}^{k} (a_{i\ell}x_i + b_{i\ell}y_i)^{d_i}. \tag{2}$$

The blocks (x_i, y_i) and their degrees d_i are fixed throughout.

Canino, Casarotti and Santarsiero [3, Theorem 1.1] give an explicit decomposition proving the general upper bound

$$R_{\mathbf{d}}(F) \leq M, \qquad M := 2^{k-1}\left(\sum_{i=1}^{k} d_i - 2k + 2\right), \qquad k \geq 2, \quad d_i \geq 3. \tag{3}$$

Our contribution is the matching lower bound for every decomposition of the form (2), without any restriction on its support. This is the partially symmetric rank question in problem TR-29 of the Open Problems in Numerical Linear Algebra catalogue [5].

*Corresponding author.

Theorem 1.1 (Exact partially symmetric rank)**.** *Let $k \geq 2$ and $d_i \geq 3$ for every i. Over $\mathbb{C}$,*

$$R_{\mathbf{d}}\left(\prod_{i=1}^{k} x_i^{d_i-1} y_i\right) = 2^{k-1}\left(\sum_{i=1}^{k} d_i - 2k + 2\right). \tag{4}$$

The proof of the lower bound is developed in Sections 2–7 and completed at the end of Section 7. The upper bound is (3).

Example 1.1 (Three cubic factors)**.** For $F = x_1^2 y_1 x_2^2 y_2 x_3^2 y_3$, the theorem gives $R_{(3,3,3)}(F) = 20$. More generally, $R_{(3,\dots,3)}(F) = 2^{k-1}(k+2)$.

The proof follows a concrete sequence. An arbitrary r-term decomposition gives r weighted points after a change of coordinates. Coefficient matching determines their moments. We then construct M test functions and prove that their evaluation vectors are linearly independent by showing that a weighted Gram matrix is invertible. Consequently $r \geq M$. The test functions and the matrix blocks are given explicitly in Section 7; the longer coefficient calculations are collected in the appendices.

All ranks in Theorem 1.1 refer to the prescribed partially symmetric blocks. The argument does not assert the same lower bound for ordinary tensor rank or border rank. All pairings below are complex bilinear, using transpose rather than conjugate transpose.

Section 8 extends the lower-bound argument to all products of binary monomials $\prod_i x_i^{a_i} y_i^{b_i}$ with $a_i \geq b_i \geq 1$ and determines their exact partially symmetric rank, together with a matching explicit decomposition.

2 A change of coordinates

Fix any decomposition (2) and delete its zero summands; this does not increase r. We first replace each pair of coefficients $(a_{i\ell}, b_{i\ell})$ by one affine parameter, absorbing its scale into the weight of the summand.

Lemma 2.1 (Affine normalization)**.** *There exist $\gamma_1, \dots, \gamma_k \in \mathbb{C}$ such that the invertible coordinate changes $x_i = X_i$, $y_i = \gamma_i X_i + Y_i$ transform (2) into the equivalent identity*

$$\prod_{i=1}^{k} X_i^{d_i-1}(\gamma_i X_i + Y_i) = \sum_{\ell=1}^{r} w_\ell \prod_{i=1}^{k} (X_i + z_{i\ell} Y_i)^{d_i}, \tag{5}$$

where every w_ℓ is nonzero and

$$z_{i\ell} = \frac{b_{i\ell}}{a_{i\ell} + \gamma_i b_{i\ell}}, \qquad w_\ell = c_\ell \prod_{i=1}^{k} (a_{i\ell} + \gamma_i b_{i\ell})^{d_i}. \tag{6}$$

Proof. For each i, choose γ_i so that $a_{i\ell} + \gamma_i b_{i\ell} \neq 0$ for every ℓ. Because zero summands were deleted, every $c_\ell \neq 0$ and every pair $(a_{i\ell}, b_{i\ell}) \neq (0,0)$. Such a choice is therefore possible: a pair with $b_{i\ell} \neq 0$ excludes only $-a_{i\ell}/b_{i\ell}$, and a pair with $b_{i\ell} = 0$ has $a_{i\ell} \neq 0$ and excludes nothing. After the coordinate change, $a_{i\ell} x_i + b_{i\ell} y_i = (a_{i\ell} + \gamma_i b_{i\ell})(X_i + z_{i\ell} Y_i)$. Substitution gives (5) and (6); hence every $w_\ell \neq 0$. □

Arrange the parameters into the matrix

$$Z_c = \begin{pmatrix} z_{11} & z_{12} & \cdots & z_{1r} \\ z_{21} & z_{22} & \cdots & z_{2r} \\ \vdots & \vdots & & \vdots \\ z_{k1} & z_{k2} & \cdots & z_{kr} \end{pmatrix} \in \mathbb{C}^{k \times r}, \qquad z_\ell = (z_{1\ell}, \dots, z_{k\ell}) \in \mathbb{C}^k. \tag{7}$$

Its ℓth column specifies the point z_ℓ. Thus the original decomposition determines r points with weights w_ℓ. These points are fixed in the subsequent argument; we choose functions to evaluate at them. Repeated points and zero coordinates are allowed. The symbol $Z = (Z_1, \dots, Z_k)$ will denote formal variables, distinct from the parameter matrix Z_c.

3 Coefficient matching and moment conditions

Lemma 3.1 (Moment conditions)**.** *The normalized identity* (5) *is equivalent to*

$$\sum_{\ell=1}^{r} w_\ell \prod_{i=1}^{k} z_{i\ell}^{e_i} = \prod_{i=1}^{k} m_i(e_i), \qquad 0 \le e_i \le d_i, \tag{8}$$

where

$$m_i(e) = \begin{cases} \gamma_i, & e = 0, \\ \beta_i := 1/d_i, & e = 1, \\ 0, & 2 \le e \le d_i. \end{cases} \tag{9}$$

Proof. Compare the coefficients of $\prod_i X_i^{d_i - e_i} Y_i^{e_i}$ on both sides of (5) and divide both by $\prod_i \binom{d_i}{e_i}$. On the right, the factor $(X_i + z_{i\ell} Y_i)^{d_i}$ contributes $\binom{d_i}{e_i} z_{i\ell}^{e_i}$. The coefficient contributed by the ith factor on the left is γ_i when $e_i = 0$, is 1 when $e_i = 1$, and is zero otherwise; after division by $\binom{d_i}{e_i}$ these are γ_i, $\beta_i = 1/d_i$ and 0. This gives (8)–(9). Conversely, these identities match every coefficient of the two multihomogeneous polynomials, so they imply (5). □

The term *moment* refers to the weighted monomial evaluations on the left of (8). The weights are complex; they are not assumed positive or normalized to sum to one. In particular, the moment conditions retain all the information in the normalized polynomial identity, rather than providing only necessary tests.

4 Generalized moment conditions

Let $\mathcal{P}_{\mathbf{d}} := \{P \in \mathbb{C}[Z_1, \dots, Z_k] : \deg_{Z_i} P \le d_i \text{ for all } i\}$. Define the weighted evaluation functional

$$L(P) := \sum_{\ell=1}^{r} w_\ell P(z_\ell), \qquad P \in \mathcal{P}_{\mathbf{d}}. \tag{10}$$

For each i, let L_i be the linear functional on univariate polynomials of degree at most d_i determined by $L_i(Z_i^e) = m_i(e)$.

Lemma 4.1 (Generalized moment conditions)**.** *The moment conditions* (8) *are equivalent to*

$$L(P) = \sum_{e} p_e \prod_{i=1}^{k} m_i(e_i) \quad \textit{for every } P(Z) = \sum_{e} p_e \prod_{i=1}^{k} Z_i^{e_i} \in \mathcal{P}_{\mathbf{d}}. \tag{11}$$

In particular, for $P(Z) = \prod_i P_i(Z_i)$ *with* $\deg P_i \le d_i$,

$$L\left(\prod_{i=1}^{k} P_i(Z_i)\right) = \prod_{i=1}^{k} L_i(P_i). \tag{12}$$

Proof. Expand P in the monomial basis and apply (8) term by term. Conversely, taking $P(Z) = \prod_i Z_i^{e_i}$ in (11) recovers each moment condition. Expanding the univariate factors gives (12). □

Thus the normalized decomposition, the monomial moment conditions, and the generalized moment conditions are equivalent descriptions of the same identity. Formula (11) determines $L(P)$ from the target polynomial whenever P lies in the specified degree range. We also use $L(R) = \sum_\ell w_\ell R(z_\ell)$ for rational functions R defined at every point, but (11) does not directly apply to such functions.

5 Constructing the weighted Gram matrix

We now select a family of expressions to which weighted evaluation will be applied. Let $t = (t_1, \dots, t_k) \in \mathbb{C}^k$ satisfy $1 + t_i z_{i\ell} \ne 0$ for every i, ℓ (we call such t admissible), and set

$$h_i(Z_i) := 1 + t_i Z_i, \qquad H_t(Z) := \prod_{i=1}^{k} h_i(Z_i)^{d_i}. \tag{13}$$

Choose M rational test functions $\phi_1, \dots, \phi_M$ defined at the points z_ℓ. Their specific choice is given in Section 7. For each $a, b = 1, \dots, M$, put

$$P_{ab}(Z) := H_t(Z)\phi_a(Z)\phi_b(Z), \tag{14}$$

and define the Gram coefficient

$$\mathcal{G}_{ab} := L(P_{ab}) = \sum_{\ell=1}^{r} w_\ell H_t(z_\ell)\phi_a(z_\ell)\phi_b(z_\ell). \tag{15}$$

If $P_{ab} \in \mathcal{P}_{\mathbf{d}}$, this coefficient can be evaluated directly by (11). Among the entries outside $\mathcal{P}_{\mathbf{d}}$, only those with $s + s' = d_i + 1$ in Lemma 7.2 are needed; they are handled by an exact rational identity in Appendix B. Entries with $s + s' \geq d_i + 2$ are never evaluated.

Lemma 5.1 (Gram factorization). *The matrix $\mathcal{G} = (\mathcal{G}_{ab})_{a,b=1}^{M}$ satisfies*

$$\mathcal{G} = E^{\mathsf{T}} W E, \tag{16}$$

where

$$E = \begin{pmatrix} \phi_1(z_1) & \phi_2(z_1) & \cdots & \phi_M(z_1) \\ \phi_1(z_2) & \phi_2(z_2) & \cdots & \phi_M(z_2) \\ \vdots & \vdots & & \vdots \\ \phi_1(z_r) & \phi_2(z_r) & \cdots & \phi_M(z_r) \end{pmatrix} \in \mathbb{C}^{r \times M}, \tag{17}$$

and

$$W = \begin{pmatrix} w_1 H_t(z_1) & 0 & \cdots & 0 \\ 0 & w_2 H_t(z_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_r H_t(z_r) \end{pmatrix} \in \mathbb{C}^{r \times r}. \tag{18}$$

Proof. The (a, b) entry of $E^{\mathsf{T}} W E$ is exactly the sum in (15). □

Each column of E is the evaluation vector $\mathrm{ev}(\phi_a) = (\phi_a(z_1), \dots, \phi_a(z_r))^{\mathsf{T}}$. The indices a and b select two functions from the same list. They do not refer to two different types of test.

6 From Gram invertibility to a rank lower bound

The matrix $\mathcal{G}$ is the weighted bilinear Gram matrix of the evaluation vectors. Its invertibility gives the required lower bound on the number of decomposition terms.

Lemma 6.1 (The dimension bound). *For the matrices in Lemma 5.1,*

$$\mathcal{G} \text{ invertible} \quad \Longrightarrow \quad \operatorname{rank} E = M \quad \Longrightarrow \quad r \geq M. \tag{19}$$

Proof. If $Ec = 0$, then $\mathcal{G}c = E^{\mathsf{T}} W E c = 0$. Invertibility of $\mathcal{G}$ gives $c = 0$, so the M columns of E are linearly independent in $\mathbb{C}^r$. □

This implication requires neither a positive weight matrix nor a Hermitian inner product. It remains to choose the M test functions and prove that their Gram matrix is invertible for at least one admissible finite parameter vector t.

7 A specific family of test functions

We now fix the test functions. All remaining assertions concern this specific choice. Write $[k] = \{1, \dots, k\}$, and use ι for a multi-index with entries in $\{0, 1\}$.

7.1 The function list and its evaluation matrix

The basic family is

$$\mathscr{B}_0 := \left\{ \prod_{j=1}^{k} \frac{Z_j^{\iota_j}}{h_j(Z_j)} : \iota \in \{0,1\}^k \right\}. \tag{20}$$

For each $i \in [k]$, add the family

$$\mathscr{B}_i := \left\{ \frac{Z_i^s}{h_i(Z_i)^s} \prod_{j \neq i} \frac{Z_j^{\iota_j}}{h_j(Z_j)} : 2 \le s \le d_i - 1, \quad \iota \in \{0,1\}^{[k]\setminus\{i\}} \right\}. \tag{21}$$

List $\mathscr{B}_0, \mathscr{B}_1, \dots, \mathscr{B}_k$ in this order. Within $\mathscr{B}_i$, order first by increasing s, then lexicographically by ι; order $\mathscr{B}_0$ lexicographically. Denote the resulting list by $\phi_1, \dots, \phi_M$. Its length is

$$M = 2^k + \sum_{i=1}^{k} (d_i - 2) 2^{k-1} = 2^{k-1} \left(\sum_{i=1}^{k} d_i - 2k + 2 \right). \tag{22}$$

This is a count of the indexed list; the linear independence of the evaluation vectors $\mathrm{ev}(\phi_a)$ is proved below. Both ϕ_a and ϕ_b in (14) are chosen from this list, possibly from the same family.

Example 7.1 (The twenty functions for three cubic factors)**.** When $k = 3$ and $d_1 = d_2 = d_3 = 3$, let $h_i = 1 + t_i Z_i$ and $h = h_1 h_2 h_3$. With the above ordering, the four rows below list the four families:

$$\begin{aligned}
(\phi_1, \dots, \phi_8) &= \frac{1}{h}(1, Z_3, Z_2, Z_2Z_3, Z_1, Z_1Z_3, Z_1Z_2, Z_1Z_2Z_3), \\
(\phi_9, \dots, \phi_{12}) &= \frac{Z_1^2}{h_1^2 h_2 h_3}(1, Z_3, Z_2, Z_2Z_3), \\
(\phi_{13}, \dots, \phi_{16}) &= \frac{Z_2^2}{h_1 h_2^2 h_3}(1, Z_3, Z_1, Z_1Z_3), \\
(\phi_{17}, \dots, \phi_{20}) &= \frac{Z_3^2}{h_1 h_2 h_3^2}(1, Z_2, Z_1, Z_1Z_2).
\end{aligned} \tag{23}$$

Their evaluations form

$$E = \big[E_0 \mid E_1 \mid E_2 \mid E_3 \big], \qquad E_0 \in \mathbb{C}^{r \times 8}, \quad E_i \in \mathbb{C}^{r \times 4} \ (i = 1, 2, 3). \tag{24}$$

For example, the ℓth row of E_1 is $\frac{z_{1\ell}^2}{h_1(z_{1\ell})^2 h_2(z_{2\ell}) h_3(z_{3\ell})}(1, z_{3\ell}, z_{2\ell}, z_{2\ell}z_{3\ell})$.

7.2 The basic block and the vanishing cross blocks

Partition the Gram matrix according to the function families. Let $\mathcal{G}_0$ be its block on $\mathscr{B}_0$, and let $\mathcal{G}_i$ be its block on $\mathscr{B}_i$ for $i \ge 1$.

Lemma 7.1 (Block diagonal structure)**.** *For every admissible t, the Gram matrix of (20)–(21) has the form*

$$\mathcal{G} = \begin{pmatrix} \mathcal{G}_0 & 0 & \cdots & 0 \\ 0 & \mathcal{G}_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathcal{G}_k \end{pmatrix}, \qquad \mathcal{G}_0 = \bigotimes_{i=1}^{k} G_i(t_i), \tag{25}$$

where

$$G_i(t_i) = \begin{pmatrix} \gamma_i + (d_i - 2) t_i \beta_i & \beta_i \\ \beta_i & 0 \end{pmatrix}, \qquad \det G_i = -\beta_i^2 \neq 0. \tag{26}$$

In particular, $\mathcal{G}_0$ is invertible for every admissible t.

Proof. For two basic functions with indices $\iota, \jmath$, multiplication by H_t cancels the denominators and gives

$$P_{ab}(Z) = \prod_i Z_i^{\iota_i + \jmath_i} h_i(Z_i)^{d_i - 2} \in \mathcal{P}_{\mathbf{d}}. \tag{27}$$

The product rule (12) shows that the Gram entry factors over the coordinates. For each coordinate the three possibilities $\iota_i + \jmath_i = 0, 1, 2$ give, respectively, $\gamma_i + (d_i - 2)t_i\beta_i$, β_i, and 0. This proves the Kronecker product formula and (26).

For a function from $\mathscr{B}_i$ paired with one from $\mathscr{B}_0$, the ith coordinate factor of P_{ab} is $Z_i^{s+\jmath_i} h_i(Z_i)^{d_i-s-1}$. It is a polynomial whose monomials have degrees between 2 and d_i, so its L_i-value is zero. All other factors lie within their prescribed degree bounds. The same factor occurs when a function from $\mathscr{B}_i$ is paired with one from $\mathscr{B}_j$, $j \neq i$, and the jth factor $Z_j^{s'+\iota_j} h_j(Z_j)^{d_j-s'-1}$ is a polynomial of degree at most d_j (see (61)). Hence all cross blocks vanish. The full entry formulas are recorded in Appendix A. □

Example 7.2 (An explicit basic Gram block)**.** For $k = 2$, write $A_i = \gamma_i + (d_i - 2)t_i\beta_i$. In the order $(0,0), (0,1), (1,0), (1,1)$,

$$G_1 \otimes G_2 = \begin{pmatrix} A_1A_2 & A_1\beta_2 & \beta_1A_2 & \beta_1\beta_2 \\ A_1\beta_2 & 0 & \beta_1\beta_2 & 0 \\ \beta_1A_2 & \beta_1\beta_2 & 0 & 0 \\ \beta_1\beta_2 & 0 & 0 & 0 \end{pmatrix}. \tag{28}$$

The product structure comes from (12), even though the points z_ℓ need not form a Cartesian grid.

7.3 The additional blocks

Put $q = 2^{k-1}$. Inside $\mathcal{G}_i$, group the functions by $s = 2, \dots, d_i - 1$. Each (s, s') block is a $q \times q$ matrix whose rows and columns are indexed by $\iota, \jmath$ on $[k] \setminus \{i\}$. Define the $q \times q$ matrix $H_i(t)$ by

$$[H_i(t)]_{\iota,\jmath} = \sum_{\ell=1}^{r} w_\ell \frac{z_{i\ell}^{d_i+1}}{1 + t_i z_{i\ell}} \prod_{j \neq i} z_{j\ell}^{\iota_j+\jmath_j} (1 + t_j z_{j\ell})^{d_j-2}. \tag{29}$$

Here H_i is a matrix; it is distinct from the scalar polynomial H_t in (13).

Lemma 7.2 (The anti-triangular blocks)**.** *The (s, s') block of $\mathcal{G}_i$ is zero if $s + s' \leq d_i$ and equals $H_i(t)$ if $s + s' = d_i + 1$. Consequently,*

$$\det \mathcal{G}_i = \pm \det H_i(t)^{d_i-2}. \tag{30}$$

Proof. The relevant entry is

$$\sum_{\ell=1}^{r} w_\ell z_{i\ell}^{s+s'} (1 + t_i z_{i\ell})^{d_i-s-s'} \prod_{j \neq i} z_{j\ell}^{\iota_j+\jmath_j} (1 + t_j z_{j\ell})^{d_j-2}. \tag{31}$$

If $s + s' \leq d_i$, the expression inside the weighted sum is a polynomial in $\mathcal{P}_{\mathbf{d}}$ whose ith-coordinate factor has all degrees at least 4. Its moment therefore vanishes. If $s + s' = d_i + 1$, the entry is (29). Reversing the order of the s'-block columns gives a block lower triangular matrix with $d_i - 2$ copies of H_i on the diagonal. This proves (30). □

Example 7.3 (Column reversal when $d_i = 5$)**.** There are three groups, indexed by $s = 2, 3, 4$. Writing J_i, K_i for the remaining $q \times q$ blocks, the matrices before and after block-column reversal are

$$\mathcal{G}_i = \begin{pmatrix} 0 & 0 & H_i \\ 0 & H_i & J_i \\ H_i & J_i & K_i \end{pmatrix} \longmapsto \begin{pmatrix} H_i & 0 & 0 \\ J_i & H_i & 0 \\ K_i & J_i & H_i \end{pmatrix}. \tag{32}$$

Thus only the invertibility of H_i is needed; no formula for J_i or K_i is required.

7.4 A common finite parameter choice

The remaining task is to show that all the matrices H_i can be made invertible at the same admissible parameter vector.

Lemma 7.3 (Nonzero determinant of each additional block)**.** *For every i, $\det H_i(t)$ is a nonzero rational function of t. More precisely, for arbitrary fixed t_j $(j \neq i)$, its matrix has the Laurent expansion at $t_i = \infty$*

$$H_i(t) = (-1)^{d_i-1}\beta_i t_i^{-d_i} \bigotimes_{j\neq i} G_j(t_j) + O(t_i^{-d_i-1}). \tag{33}$$

The leading coefficient matrix is invertible.

Proof. Expand $z^{d_i+1}/(1+t_iz)$ through order $t_i^{-d_i}$ in (29). The coefficients involving $z_i^{d_i}, \dots, z_i^2$ vanish by the moment conditions. The next coefficient involves z_i; the product rule gives $\beta_i \bigotimes_{j\neq i} G_j(t_j)$, with the sign in (33). Appendix B gives the exact finite identity and verifies this calculation also when some $z_{i\ell} = 0$. Each G_j is invertible by Lemma 7.1, so the leading coefficient is invertible and $\det H_i$ is not identically zero. □

Proposition 7.1 (Simultaneous Gram nondegeneracy)**.** *For any points and weights satisfying (8), there exists a finite $t \in \mathbb{C}^k$ such that $1 + t_iz_{i\ell} \neq 0$ for every i, ℓ and the Gram matrix of (20)–(21) is invertible.*

Proof. Put $D_i := \prod_{\ell=1}^{r}(1+t_iz_{i\ell})^q$ and $N_i := D_i \det H_i$. The entries of H_i have denominators dividing $\prod_\ell(1+t_iz_{i\ell})$, so $N_i \in \mathbb{C}[t]$; both N_i and D_i are nonzero polynomials, and $\det H_i = N_i/D_i$. The polynomial

$$Q(t) := \left(\prod_{i,\ell}(1+t_iz_{i\ell})\right)\prod_{i=1}^{k} N_i(t)D_i(t) \tag{34}$$

is nonzero. Since $\mathbb{C}$ is infinite, there is a finite parameter vector with $Q(t) \neq 0$. At this vector all evaluations are defined and all H_i are invertible. By Lemma 7.2, each $\mathcal{G}_i$ is invertible; by Lemma 7.1, the full block diagonal matrix $\mathcal{G}$ is invertible. □

The one-parameter expansions in Lemma 7.3 prove that the individual determinants are nonzero rational functions. They are not used to take all parameters to infinity simultaneously. The finite choice in (34) supplies the single parameter vector needed in the dimension argument.

Example 7.4 (The 20×20 Gram matrix)**.** For the cubic triple in Example 7.1, each additional family has only $s = 2$, so $\mathcal{G}_i = H_i$ for $i = 1, 2, 3$. The full matrix is

$$\mathcal{G} = \begin{pmatrix} G_1\otimes G_2\otimes G_3 & 0_{8\times4} & 0_{8\times4} & 0_{8\times4} \\ 0_{4\times8} & H_1 & 0_{4\times4} & 0_{4\times4} \\ 0_{4\times8} & 0_{4\times4} & H_2 & 0_{4\times4} \\ 0_{4\times8} & 0_{4\times4} & 0_{4\times4} & H_3 \end{pmatrix}. \tag{35}$$

Here $G_i = \left(\begin{smallmatrix} \gamma_i+t_i/3 & 1/3 \\ 1/3 & 0 \end{smallmatrix}\right)$ and each H_i is 4×4. For example, $H_1 = \frac{1}{3}t_1^{-3}(G_2\otimes G_3) + O(t_1^{-4})$ with t_2, t_3 fixed. The other two blocks have the analogous expansions. Proposition 7.1 makes all four diagonal blocks invertible at one finite t. Thus the twenty columns of E are independent.

Proof of Theorem 1.1. Take an arbitrary decomposition (2) with r nonzero terms. By Lemmas 2.1 and 3.1, it determines points and nonzero weights satisfying (8) with $\beta_i = 1/d_i$. Construct the M test functions (20)–(21) and choose t using Proposition 7.1. Their Gram matrix is invertible, so Lemma 6.1 yields $r \geq M$. This applies to every decomposition, hence $R_{\mathbf{d}}(F) \geq M$. This lower bound uses neither $k \geq 2$ nor $d_i \geq 3$; it holds for all $k \geq 1$ and $d_i \geq 2$. Combining with (3) proves (4). □

8 Products of binary monomials

The method of Sections 2–7 extends from $x_i^{d_i-1}y_i$ to binary monomials $x_i^{a_i}y_i^{b_i}$ with $a_i \geq b_i \geq 1$. Let $\mathbf{a} = (a_1, \dots, a_k)$ and $\mathbf{b} = (b_1, \dots, b_k)$ be integer vectors with $a_i \geq b_i \geq 1$, put $d_i = a_i + b_i$, and consider

$$F_{\mathbf{a},\mathbf{b}}(x, y) = \prod_{i=1}^{k} x_i^{a_i}y_i^{b_i}. \tag{36}$$

Its partially symmetric rank $R_{\mathbf{d}}(F_{\mathbf{a},\mathbf{b}})$ is the least r for which $F_{\mathbf{a},\mathbf{b}}$ admits an expression of the form (2), with the blocks (x_i, y_i) and the degrees d_i fixed. Interchanging x_i and y_i preserves the form; for arbitrary positive exponents the rank is obtained by replacing (a_i, b_i) with $(\max\{a_i, b_i\}, \min\{a_i, b_i\})$. When every $b_i = 1$, $F_{\mathbf{a},\mathbf{b}}$ is the polynomial (1).

Theorem 8.1 (Products of binary monomials)**.** *Let $k \geq 1$, let $a_i \geq b_i \geq 1$ be integers, and put $d_i = a_i + b_i$. Over $\mathbb{C}$,*

$$R_{\mathbf{d}}\left(\prod_{i=1}^{k} x_i^{a_i} y_i^{b_i}\right) = N(\mathbf{a},\mathbf{b}) := \prod_{i=1}^{k}(b_i+1) + \sum_{i=1}^{k}(a_i - b_i)\prod_{j\neq i}(b_j+1), \tag{37}$$

where empty products equal 1*.*

The lower bound holds for every finite decomposition (2) of $F_{\mathbf{a},\mathbf{b}}$, without restrictions on its complex coefficients or its support. The upper bound is realized by the explicit identity (58). The theorem concerns the exact complex rank with the symmetric blocks fixed. It makes no assertion about real rank, border rank, cactus rank, unrestricted nonsymmetric rank, or rank after grouping modes from different blocks, and the lower-bound argument uses only finite decompositions. Pure-power factors, with $b_i = 0$, are excluded; see Remark 8.2.

The lower bound is first proved for abstract product moment conditions in Theorem 8.2, and Proposition 8.1 reduces an arbitrary decomposition to such conditions. The upper bound is the finite-character decomposition of Proposition 8.2. In this section, $m_i(e)$ denotes general moment values as in Theorem 8.2; the values (9) are a special case.

8.1 A product-moment lower bound

Throughout this subsection, $b_i \geq 1$ and $d_i \geq 2b_i$ are integers for $i \in [k]$, and

$$\delta_i := d_i - 2b_i, \qquad q_i := \prod_{j\neq i}(b_j+1), \qquad N := \prod_{i=1}^{k}(b_i+1) + \sum_{i=1}^{k}\delta_i q_i. \tag{38}$$

When every $b_j = 1$, these are $\delta_i = d_i - 2$, $q_i = q = 2^{k-1}$ and $N = M$; see (22).

Theorem 8.2 (Product-moment lower bound)**.** *Let $k \geq 1$, let $b_i \geq 1$ and $d_i \geq 2b_i$ be integers, and let N be as in (38). For each i, let $m_i(0), \dots, m_i(d_i) \in \mathbb{C}$ satisfy*

$$m_i(e) = 0 \quad (b_i < e \leq d_i), \qquad \beta_i := m_i(b_i) \neq 0. \tag{39}$$

Let $z_1, \dots, z_r \in \mathbb{C}^k$ and nonzero $w_1, \dots, w_r \in \mathbb{C}$ satisfy

$$\sum_{\ell=1}^{r} w_\ell \prod_{i=1}^{k} z_{i\ell}^{e_i} = \prod_{i=1}^{k} m_i(e_i), \qquad 0 \leq e_i \leq d_i. \tag{40}$$

Then $r \geq N$. The points may repeat, and any of their coordinates may be zero.

Fix data as in Theorem 8.2 for the rest of this subsection. Let L_i be the linear functional on univariate polynomials of degree at most d_i determined by $L_i(Z_i^e) = m_i(e)$, and let L be the weighted evaluation (10). The proof of Lemma 4.1 uses only linearity, so the lemma holds verbatim with (40) in place of (8). In particular, the product rule (12) holds whenever $\deg P_i \leq d_i$ for every i; all Gram computations below use (40) only in this form.

Let $t \in \mathbb{C}^k$ be admissible, that is, $1 + t_i z_{i\ell} \neq 0$ for every i, ℓ, and let h_i and H_t be as in (13). For rational functions ϕ, ψ defined at every point z_ℓ, put

$$\langle \phi, \psi \rangle_t := \sum_{\ell=1}^{r} w_\ell H_t(z_\ell)\phi(z_\ell)\psi(z_\ell). \tag{41}$$

This is a symmetric bilinear form; no positivity is involved. The Gram matrix of a finite list of such functions is the matrix (15) of that list, so Lemmas 5.1 and 6.1 apply to it in place of $\phi_1, \dots, \phi_M$. The identities below hold for every admissible t; equivalently, they hold over the field $\mathbb{C}(t_1, \dots, t_k)$ with indeterminate parameters, in which every $1 + t_i z_{i\ell}$ is a nonzero polynomial, also when $z_{i\ell} = 0$.

Write $\mathcal{I} := \prod_{j=1}^{k}\{0, \dots, b_j\}$ and $\mathcal{I}_i := \prod_{j\neq i}\{0, \dots, b_j\}$. Empty products, including empty Kronecker products of matrices, equal 1 (the 1×1 matrix (1)); when $k = 1$, $\mathcal{I}_1$ consists of the empty tuple. The basic list is

$$\mathscr{U} := (f_\iota)_{\iota\in\mathcal{I}}, \qquad f_\iota(Z) := \prod_{j=1}^{k} \frac{Z_j^{\iota_j}}{h_j(Z_j)^{b_j}}, \tag{42}$$

ordered lexicographically. For each i with $\delta_i > 0$, the additional list $\mathscr{V}_i$ consists of the functions

$$g_{i,s,\iota}(Z) := \frac{Z_i^s}{h_i(Z_i)^s} \prod_{j \neq i} \frac{Z_j^{\iota_j}}{h_j(Z_j)^{b_j}}, \qquad b_i + 1 \le s \le d_i - b_i, \quad \iota \in \mathcal{I}_i, \tag{43}$$

ordered first by increasing s and then lexicographically by ι; if $\delta_i = 0$, then $\mathscr{V}_i$ is empty. Thus $\mathscr{U}$ has $\prod_j (b_j + 1)$ members and $\mathscr{V}_i$ has $\delta_i q_i$ members, N in total. When every $b_j = 1$, $\mathscr{U}$ and $\mathscr{V}_i$ are the families $\mathscr{B}_0$ and $\mathscr{B}_i$ of (20)–(21).

For $0 \le e, e' \le b_i$, define the $(b_i + 1) \times (b_i + 1)$ matrix $G_i(t_i)$ by

$$[G_i(t_i)]_{e,e'} := L_i\Big(Z_i^{e+e'} h_i(Z_i)^{\delta_i} \Big). \tag{44}$$

Lemma 8.1 (The basic block)**.** *For every admissible t, the Gram matrix of $\mathscr{U}$ is $\bigotimes_{i=1}^k G_i(t_i)$, and*

$$\det G_i(t_i) = (-1)^{b_i(b_i+1)/2} \beta_i^{b_i+1} \neq 0. \tag{45}$$

In particular, the Gram matrix of $\mathscr{U}$ is invertible for every admissible t.

Proof. The polynomial in (44) has degree at most $2b_i + \delta_i = d_i$. If $e + e' > b_i$, each of its monomials has degree between $b_i + 1$ and d_i, so the entry vanishes by (39). If $e + e' = b_i$, only the lowest term survives, and the entry equals β_i. Reversing the order of the columns therefore gives an upper triangular matrix whose diagonal entries all equal β_i; the reversal of $b_i + 1$ columns has sign $(-1)^{b_i(b_i+1)/2}$, which proves (45). Finally, for $\iota, \jmath \in \mathcal{I}$, multiplication by H_t cancels the denominators, and the product rule gives

$$\langle f_\iota, f_\jmath \rangle_t = L\left(\prod_{i=1}^k Z_i^{\iota_i + \jmath_i} h_i(Z_i)^{\delta_i} \right) = \prod_{i=1}^k [G_i(t_i)]_{\iota_i, \jmath_i},$$

since the ith factor has degree at most d_i. This is the $(\iota, \jmath)$ entry of the Kronecker product. □

Lemma 8.2 (Orthogonality of the lists)**.** *For every admissible t, the lists $\mathscr{U}, \mathscr{V}_1, \dots, \mathscr{V}_k$ are mutually orthogonal for (41).*

Proof. Pair $g_{i,s,\iota}$ with $f_\jmath$. After multiplication by H_t, the factor in coordinate i is

$$Z_i^{s+\jmath_i} h_i(Z_i)^{d_i - s - b_i}.$$

The exponent of h_i is nonnegative because $s \le d_i - b_i$. Every monomial has degree at least $s > b_i$ and at most $d_i - b_i + \jmath_i \le d_i$, so its L_i-value is zero. Every other coordinate factor is $Z_j^{\iota_j + \jmath_j} h_j(Z_j)^{\delta_j}$, of degree at most d_j. The product rule gives $\langle g_{i,s,\iota}, f_\jmath \rangle_t = 0$.

Now let $i \neq j$, and pair $g_{i,s,\iota} \in \mathscr{V}_i$ with $g_{j,s',\jmath} \in \mathscr{V}_j$. The coordinate-$i$ factor has the same form $Z_i^{s+\jmath_i} h_i(Z_i)^{d_i - s - b_i}$, with $\jmath_i$ supplied by the second function, and again has zero L_i-value. The coordinate-j factor $Z_j^{s'+\iota_j} h_j(Z_j)^{d_j - s' - b_j}$ has a nonnegative exponent on h_j and degree at most d_j. All remaining coordinate factors have the form used in Lemma 8.1. Thus every use of the product rule stays inside the degree box, and $\langle g_{i,s,\iota}, g_{j,s',\jmath} \rangle_t = 0$. □

When every $b_j = 1$, these are the computations of Appendix A. It remains to treat the Gram matrix of a nonempty $\mathscr{V}_i$. Define the $q_i \times q_i$ matrix $H_i(t)$, with rows and columns indexed by $\iota, \jmath \in \mathcal{I}_i$, by

$$[H_i(t)]_{\iota,\jmath} := \sum_{\ell=1}^r w_\ell \frac{z_{i\ell}^{d_i+1}}{1 + t_i z_{i\ell}} \prod_{j \neq i} z_{j\ell}^{\iota_j + \jmath_j} (1 + t_j z_{j\ell})^{\delta_j}. \tag{46}$$

When every $b_j = 1$, this is the matrix (29).

Lemma 8.3 (The anti-triangular blocks)**.** *Let $\delta_i > 0$, and group $\mathscr{V}_i$ by s. For every admissible t, the (s, s') block of the Gram matrix of $\mathscr{V}_i$ is zero if $s + s' \le d_i$ and equals $H_i(t)$ if $s + s' = d_i + 1$. Consequently, up to the sign of a block-column permutation, the Gram determinant of $\mathscr{V}_i$ equals $\det H_i(t)^{\delta_i}$.*

Proof. The (s, s') block has entries

$$\sum_{\ell=1}^{r} w_\ell z_{i\ell}^{s+s'} (1 + t_i z_{i\ell})^{d_i - s - s'} \prod_{j \neq i} z_{j\ell}^{\iota_j + \jmath_j} (1 + t_j z_{j\ell})^{\delta_j}. \tag{47}$$

If $s + s' \leq d_i$, this is the L-value of a polynomial in $\mathcal{P}_{\mathbf{d}}$ whose coordinate-i degrees all exceed b_i; hence the whole block is zero. If $s + s' = d_i + 1$, the block is exactly $H_i(t)$.

Write $s = b_i + 1 + \alpha$ with $0 \leq \alpha < \delta_i$. After reversing the order of the block columns, the block column with index α' corresponds to $s' = d_i - b_i - \alpha'$, so

$$s + s' = d_i + 1 + \alpha - \alpha'.$$

Every block above the diagonal ($\alpha' > \alpha$) is therefore zero, and every diagonal block is $H_i(t)$. The resulting block lower triangular matrix has determinant $\det H_i(t)^{\delta_i}$, as asserted. The blocks with $s + s' \geq d_i + 2$ lie below the diagonal and are not needed. □

The next identity shows that H_i is nonsingular as a rational matrix. Its form includes zero coordinates of the points.

Lemma 8.4 (A finite division identity)**.** *For integers $b \geq 1$ and $d \geq 2b$, the identity*

$$\frac{z^{d+1}}{1 + tz} = \sum_{e=b}^{d} (-1)^{d-e} \frac{z^e}{t^{d+1-e}} + (-1)^{d+1-b} \frac{z^b}{t^{d+1-b}(1 + tz)} \tag{48}$$

holds whenever $t \neq 0$ and $1 + tz \neq 0$. For each fixed $z \neq 0$, the final term is $O(t^{-(d-b+2)})$ as $t \to \infty$; for $z = 0$, every term of (48) vanishes identically. For $b = 1$, the substitution $n = d - e$ turns (48) into (63).

Proof. Multiply (48) by $(1+tz)t^{d+1-b}$. Consecutive terms of the finite sum cancel, leaving $t^{d+1-b} z^{d+1}$. If $z \neq 0$, the additional denominator $1 + tz$ gives one further power of t^{-1} at infinity. If $z = 0$, the factor z^b vanishes because $b \geq 1$, as do all the other numerators. Thus neither the identity nor the remainder statement requires division by z. □

Lemma 8.5 (Nonsingularity of the additional blocks)**.** *Let $\delta_i > 0$. Then $\det H_i(t)$ is a nonzero element of $\mathbb{C}(t_1, \ldots, t_k)$. More precisely, as a Laurent expansion in t_i^{-1} with coefficients in $\mathbb{C}(t_j : j \neq i)$,*

$$H_i(t) = (-1)^{d_i - b_i} \beta_i \, t_i^{-(d_i - b_i + 1)} \bigotimes_{j \neq i} G_j(t_j) + O\Big(t_i^{-(d_i - b_i + 2)}\Big), \tag{49}$$

and the leading coefficient matrix is invertible.

Proof. Treat the parameters t_j, $j \neq i$, as elements of the coefficient field $\mathbb{C}(t_j : j \neq i)$. For $\iota, \jmath \in \mathcal{I}_i$, put $A_{\iota,\jmath}(Z_{-i}) := \prod_{j \neq i} Z_j^{\iota_j + \jmath_j} h_j(Z_j)^{\delta_j}$; its degree in each Z_j is at most d_j. Applying (48) with $d = d_i$ and $b = b_i$ separately at each point in (46) gives

$$[H_i(t)]_{\iota,\jmath} = \sum_{e=b_i}^{d_i} \frac{(-1)^{d_i - e}}{t_i^{d_i + 1 - e}} L(Z_i^e A_{\iota,\jmath}) + \frac{(-1)^{d_i + 1 - b_i}}{t_i^{d_i + 1 - b_i}} \sum_{\ell=1}^{r} w_\ell \frac{z_{i\ell}^{b_i} A_{\iota,\jmath}(z_\ell)}{1 + t_i z_{i\ell}}. \tag{50}$$

The sum over the points is finite and its terms with $z_{i\ell} = 0$ vanish identically, so by Lemma 8.4 the last term is $O(t_i^{-(d_i - b_i + 2)})$ entrywise. For $b_i < e \leq d_i$, the product rule and (39) give $L(Z_i^e A_{\iota,\jmath}) = 0$. For $e = b_i$, the product rule gives

$$L\Big(Z_i^{b_i} A_{\iota,\jmath}\Big) = \beta_i \prod_{j \neq i} L_j\Big(Z_j^{\iota_j + \jmath_j} h_j(Z_j)^{\delta_j}\Big) = \beta_i \left[\bigotimes_{j \neq i} G_j(t_j)\right]_{\iota,\jmath}.$$

This proves (49), which may equivalently be read as a formal Laurent expansion in t_i^{-1}. By Lemma 8.1, the leading coefficient matrix $C_i := (-1)^{d_i - b_i} \beta_i \bigotimes_{j \neq i} G_j(t_j)$ is invertible. The coefficient of $t_i^{-(d_i - b_i + 1) q_i}$ in $\det H_i$ is therefore $\det C_i \neq 0$, so $\det H_i$ is not the zero rational function. Each coordinate i is treated separately; no common limiting path for the parameters is required. When every $b_j = 1$, (49) is (33). □

Proof of Theorem 8.2. For each i, put $D_i(t_i) := \prod_{\ell=1}^{r}(1+t_i z_{i\ell})^{q_i}$. If $\delta_i > 0$, the entries of H_i have denominators dividing $\prod_\ell(1+t_i z_{i\ell})$, so $D_i \det H_i \in \mathbb{C}[t]$; it is a nonzero polynomial by Lemma 8.5. Hence the polynomial

$$\left(\prod_{i,\ell}(1+t_i z_{i\ell})\right)\prod_{i:\,\delta_i>0} D_i(t_i)\det H_i(t) \tag{51}$$

is nonzero. Since $\mathbb{C}$ is infinite, it is nonzero at some finite $t \in \mathbb{C}^k$; fix such a t. Then t is admissible, so all the functions (42)–(43) and the pairing (41) are defined, and $\det H_i(t) \neq 0$ whenever $\delta_i > 0$. The Gram matrix of $\mathscr{U}$ is invertible by Lemma 8.1, and each nonempty $\mathscr{V}_i$ has an invertible Gram matrix by Lemma 8.3. By Lemma 8.2, the Gram matrix of the combined list $\mathscr{U}, \mathscr{V}_1, \dots, \mathscr{V}_k$ is block diagonal with these blocks, hence invertible. Lemmas 5.1 and 6.1, applied to these N functions, show that their evaluation vectors are linearly independent in $\mathbb{C}^r$, so $r \geq N$.

No point was discarded or required to be distinct or to have nonzero coordinates. If a moment formula (40) initially contains zero weights, those terms may first be deleted. □

As in Section 7, the count rests on the invertibility of every diagonal Gram block, established before dimensions are added; no positivity of the weights is used, and the auxiliary parameters are evaluated at one finite point.

8.2 From decompositions to product moments

Proposition 8.1 (Reduction to moments)**.** *Let $a_i \geq b_i \geq 1$ and $d_i = a_i + b_i$. Every decomposition (2) of $F_{\mathbf{a},\mathbf{b}}$ has at least $N(\mathbf{a},\mathbf{b})$ summands.*

Proof. To avoid a clash with the exponents a_i, b_i, write the decomposition as

$$F_{\mathbf{a},\mathbf{b}}(x,y) = \sum_{\ell=1}^{r} c_\ell \prod_{i=1}^{k}(u_{i\ell}x_i + v_{i\ell}y_i)^{d_i}, \tag{52}$$

that is, (2) with $(a_{i\ell}, b_{i\ell})$ renamed $(u_{i\ell}, v_{i\ell})$. Delete its zero summands; this does not increase r, and afterwards every $c_\ell \neq 0$ and every pair $(u_{i\ell}, v_{i\ell}) \neq (0,0)$. As in the proof of Lemma 2.1, choose $\gamma_i \in \mathbb{C}$ with $u_{i\ell} + \gamma_i v_{i\ell} \neq 0$ for every ℓ: a pair with $v_{i\ell} \neq 0$ excludes only $-u_{i\ell}/v_{i\ell}$, and a pair with $v_{i\ell} = 0$ excludes nothing. Set

$$z_{i\ell} = \frac{v_{i\ell}}{u_{i\ell} + \gamma_i v_{i\ell}}, \qquad w_\ell = c_\ell \prod_{i=1}^{k}(u_{i\ell} + \gamma_i v_{i\ell})^{d_i}, \tag{53}$$

as in (6). All weights are nonzero. The points may repeat, and $z_{i\ell}$ may be zero.

For $0 \leq e_i \leq d_i$, apply the differential operator

$$\prod_{i=1}^{k}(\partial_{x_i} + \gamma_i\partial_{y_i})^{d_i-e_i}\partial_{y_i}^{e_i}$$

to (52) and divide by $\prod_i d_i!$. Since $(\partial_{x_i} + \gamma_i\partial_{y_i})^{d_i-e_i}\partial_{y_i}^{e_i}(u_{i\ell}x_i + v_{i\ell}y_i)^{d_i} = d_i!\,(u_{i\ell} + \gamma_i v_{i\ell})^{d_i-e_i}v_{i\ell}^{e_i}$, the result on the ℓth summand is $w_\ell \prod_i z_{i\ell}^{e_i}$. On the target (36) the result factors over the blocks, and the factor of the ith block is

$$m_i(e) := \frac{1}{d_i!}(\partial_{x_i} + \gamma_i\partial_{y_i})^{d_i-e}\partial_{y_i}^{e}\left(x_i^{a_i}y_i^{b_i}\right) = \begin{cases} \dfrac{(d_i-e)!\,b_i!}{(b_i-e)!\,d_i!}\,\gamma_i^{b_i-e}, & 0 \leq e \leq b_i, \\ 0, & b_i < e \leq d_i. \end{cases} \tag{54}$$

Indeed, when $e \leq b_i$, differentiation by $\partial_{y_i}^e$ first contributes $b_i!/(b_i-e)!$. In the binomial expansion of $(\partial_{x_i} + \gamma_i\partial_{y_i})^{d_i-e}$, only the term with $b_i - e$ derivatives in y_i and a_i derivatives in x_i survives; it contributes $\binom{d_i-e}{b_i-e}\gamma_i^{b_i-e}a_i!\,(b_i-e)!$, and the product of these factors divided by $d_i!$ is (54). For $e > b_i$, the first differentiation already vanishes.

Consequently the moment conditions (40) hold throughout the entire box $0 \leq e_i \leq d_i$, with $m_i(e)$ given by (54), and

$$\beta_i = m_i(b_i) = \frac{a_i!\,b_i!}{d_i!} = \binom{d_i}{b_i}^{-1} \neq 0.$$

Because $a_i \geq b_i$, we have $d_i \geq 2b_i$ and $\delta_i = d_i - 2b_i = a_i - b_i$, so the number N of (38) is $N(\mathbf{a},\mathbf{b})$. Theorem 8.2 gives $r \geq N(\mathbf{a},\mathbf{b})$. □

The parameters γ_i and the auxiliary parameters t in Theorem 8.2 serve different purposes. The first make an arbitrary finite decomposition affine; the second make the Gram blocks nondegenerate. Both choices are made after the decomposition is fixed, and both follow from explicit finite nonvanishing conditions. Neither is a genericity assumption on the decomposition.

8.3 A finite-character decomposition

In this subsection $a_i \geq b_i \geq 1$, $d_i = a_i + b_i$, $\delta_i = a_i - b_i$ and $N = N(\mathbf{a}, \mathbf{b})$. Let $\delta = (\delta_1, \dots, \delta_k)^{\mathsf{T}}$, $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$, and let $\mathbf{e}_1, \dots, \mathbf{e}_k$ be the standard basis vectors of $\mathbb{Z}^k$. The construction uses a lattice whose nonzero vectors avoid the box of possible exponent differences, followed by character orthogonality on the corresponding finite quotient.

Lemma 8.6 (A lattice avoiding the coefficient box)**.** *Let*

$$\Delta := \operatorname{diag}(b_1 + 1, \dots, b_k + 1) + \delta \mathbf{1}^{\mathsf{T}},$$

and let $\Lambda := \Delta\mathbb{Z}^k$ be the lattice generated by the columns of Δ. Then $\det \Delta = N$, so $[\mathbb{Z}^k : \Lambda] = N$, and

$$\Lambda \cap \prod_{i=1}^{k} [-a_i, b_i] = \{0\}. \tag{55}$$

Proof. The jth column of Δ is $(b_j + 1)\mathbf{e}_j + \delta$. Expand the determinant multilinearly in these columns. Any term that chooses δ from two columns vanishes. The term choosing no such column is $\prod_i (b_i + 1)$, and the term choosing it only from column i is $\delta_i \prod_{j \neq i} (b_j + 1)$. Hence

$$\det \Delta = \prod_{i=1}^{k} (b_i + 1) + \sum_{i=1}^{k} \delta_i \prod_{j \neq i} (b_j + 1) = N > 0.$$

The index of the column lattice of a nonsingular integer matrix is the absolute value of its determinant: integer row and column operations preserve both quantities and reduce the matrix to diagonal form, where the claim is immediate. Thus $[\mathbb{Z}^k : \Lambda] = N$.

Suppose that $\lambda = \Delta\mu$ lies in the box in (55), with $\mu \in \mathbb{Z}^k$, and put $\sigma = \sum_i \mu_i$. Since Λ is generated by the columns,

$$\lambda_i = (b_i + 1)\mu_i + \delta_i \sigma. \tag{56}$$

If $\sigma > 0$, then $\lambda_i \leq b_i$ and $\delta_i \geq 0$ imply $(b_i + 1)\mu_i \leq b_i$ for every i. Thus every $\mu_i \leq 0$, contrary to $\sigma > 0$. If $\sigma < 0$, then $-\sigma - 1 \geq 0$; since $a_i = b_i + \delta_i$, the lower bound $\lambda_i \geq -a_i$ gives

$$(b_i + 1)\mu_i \geq -(b_i + 1) + 1 + (-\sigma - 1)\delta_i \geq -(b_i + 1) + 1.$$

Integrality now implies $\mu_i \geq 0$ for every i, contrary to $\sigma < 0$. Finally, if $\sigma = 0$, the upper bounds again imply $\mu_i \leq 0$ for every i, and their zero sum forces $\mu = 0$. This proves (55), including the cases in which some δ_i vanish. □

Lemma 8.7 (Finite character orthogonality)**.** *Let Γ be a finite abelian group, and let $\widehat{\Gamma} = \operatorname{Hom}(\Gamma, \mathbb{C}^\times)$. Then $|\widehat{\Gamma}| = |\Gamma|$ and, for every $\xi \in \Gamma$,*

$$\sum_{\chi \in \widehat{\Gamma}} \chi(\xi) = \begin{cases} |\Gamma|, & \xi = 0, \\ 0, & \xi \neq 0. \end{cases} \tag{57}$$

Proof. Write Γ as a product of cyclic groups $\mathbb{Z}/\nu_1\mathbb{Z} \times \cdots \times \mathbb{Z}/\nu_\kappa\mathbb{Z}$. A character is determined by independently choosing a ν_jth root of unity as the image of the jth cyclic generator. There are therefore $\prod_j \nu_j = |\Gamma|$ characters. If ξ has coordinates ξ_j, the sum in (57) factors as

$$\prod_{j=1}^{\kappa} \left(\sum_{n=0}^{\nu_j - 1} \exp\left(\frac{2\pi \mathrm{i}\, n \xi_j}{\nu_j} \right) \right).$$

Each geometric sum equals ν_j when $\xi_j = 0$ in $\mathbb{Z}/\nu_j\mathbb{Z}$ and equals zero otherwise, which proves the formula. □

Proposition 8.2 (An explicit decomposition)**.** *Let Λ be the lattice of Lemma 8.6, put $\Gamma := \mathbb{Z}^k/\Lambda$, and for $\chi \in \widehat{\Gamma}$ put $\theta_i(\chi) := \chi(\mathbf{e}_i + \Lambda)$. Then*

$$F_{\mathbf{a},\mathbf{b}}(x,y) = \frac{1}{N\prod_{i=1}^k \binom{d_i}{a_i}} \sum_{\chi\in\widehat{\Gamma}} \left(\prod_{i=1}^k \theta_i(\chi)^{-a_i}\right) \prod_{i=1}^k \big(\theta_i(\chi)x_i + y_i\big)^{d_i}. \tag{58}$$

This is a decomposition of the form (2) with N nonzero summands. In particular, $R_{\mathbf{d}}(F_{\mathbf{a},\mathbf{b}}) \le N(\mathbf{a},\mathbf{b})$.

Proof. Each $\theta_i(\chi)$ is a root of unity, and in particular nonzero. Let $\mathbf{c} = (c_1,\dots,c_k)$ satisfy $0 \le c_i \le d_i$. Since $\prod_i \theta_i(\chi)^{c_i-a_i} = \chi(\mathbf{c}-\mathbf{a}+\Lambda)$, the coefficient of $\prod_i x_i^{c_i} y_i^{d_i-c_i}$ on the right of (58) is

$$\frac{\prod_i \binom{d_i}{c_i}}{N\prod_i \binom{d_i}{a_i}} \sum_{\chi\in\widehat{\Gamma}} \chi(\mathbf{c}-\mathbf{a}+\Lambda). \tag{59}$$

Since $d_i - a_i = b_i$, the vector $\mathbf{c}-\mathbf{a}$ lies in $\prod_i[-a_i,b_i]$. By Lemma 8.6, it belongs to Λ precisely when $\mathbf{c} = \mathbf{a}$. As $|\Gamma| = N$, Lemma 8.7 therefore makes (59) zero for $\mathbf{c} \ne \mathbf{a}$ and equal to one for $\mathbf{c} = \mathbf{a}$. This verifies every coefficient of (58).

There are $|\widehat{\Gamma}| = N$ summands, and every coefficient in (58) is nonzero. The tuples $(\theta_1(\chi),\dots,\theta_k(\chi))$ are also distinct: the classes $\mathbf{e}_i + \Lambda$ generate Γ, so two characters with the same θ_i for every i coincide. Moreover, the product $\prod_i(\theta_i x_i + y_i)^{d_i}$ has coefficient one at $\prod_i y_i^{d_i}$ and coefficient $d_i\theta_i$ at $x_i y_i^{d_i-1}\prod_{j\ne i} y_j^{d_j}$; thus distinct tuples give nonproportional products. The identity is consequently an N-term decomposition over $\mathbb{C}$ with the blocks fixed. □

Proof of Theorem 8.1. Proposition 8.1 bounds every decomposition below by $N(\mathbf{a},\mathbf{b})$, and Proposition 8.2 supplies a decomposition with exactly this many summands. □

8.4 Specializations

Remark 8.1 (The case $b_i = 1$)**.** Take $b_i = 1$ and $a_i = d_i - 1$, which is allowed for every $d_i \ge 2$. Then $\delta_i = d_i - 2$ and $N(\mathbf{a},\mathbf{b}) = 2^k + \sum_i (d_i-2)2^{k-1} = M$. The moments (54) become (9), with $\beta_i = 1/d_i$; the lists $\mathscr{U}$ and $\mathscr{V}_i$ are $\mathscr{B}_0$ and $\mathscr{B}_i$; and G_i, H_i and (49) reduce to (26), (29) and (33). Thus Theorem 8.1 contains Theorem 1.1, and gives $R_{\mathbf{d}}\big(\prod_i x_i^{d_i-1}y_i\big) = 2^{k-1}(\sum_i d_i - 2k + 2)$ for all $k \ge 1$ and $d_i \ge 2$. In particular, Proposition 8.2 gives an explicit M-term decomposition for all $k \ge 1$ and $d_i \ge 2$, independently of [3, Theorem 1.1].

For one block, (37) gives $R_{(a+b)}(x^a y^b) = a + 1$ for $a \ge b \ge 1$, the classical rank of a binary monomial [1, Section 4.3 of the arXiv version]. For two blocks it becomes

$$\begin{aligned} N(\mathbf{a},\mathbf{b}) &= (b_1+1)(b_2+1) + (a_1-b_1)(b_2+1) + (a_2-b_2)(b_1+1) \\ &= (a_1+1)(a_2+1) - (a_1-b_1)(a_2-b_2). \end{aligned}$$

For two binary-monomial blocks, Gałązka constructed a reduced apolar set giving

$$R_{\mathbf{d}}\big(x_1^{a_1}y_1^{b_1}x_2^{a_2}y_2^{b_2}\big) \le (a_1+1)(b_2+1) + (b_1+1)(a_2+1) - (b_1+1)(b_2+1), \qquad a_i \ge b_i \ge 1,$$

and obtained lower bounds and exact special cases [4, Theorem 1.5 and Section 7.1 of the arXiv version]. The binomial equations in that construction describe a finite set in the two-dimensional torus. For $k = 2$ the support of the decomposition (58) is exactly the point set of Gałązka's construction [4, Section 7.1 of the arXiv version]. The lower bound

$$R_{\mathbf{d}}\big(x_1^{a_1}y_1^{b_1}x_2^{a_2}y_2^{b_2}\big) \ge \max\{(a_1+1)(b_2+1), (b_1+1)(a_2+1)\}$$

also appears in [1, Proposition 4.3 of the arXiv version] (stated there with the roles of x_i and y_i exchanged). Behr's unpublished manuscript [2] gives the exact two-block formula

$$R_{\mathbf{d}}\big(x_1^{a_1}y_1^{b_1}x_2^{a_2}y_2^{b_2}\big) = (a_1+1)(a_2+1) - (a_1-b_1)(a_2-b_2) \qquad (a_i \ge b_i \ge 1)$$

in Corollary 5.4, including the value $2(d_1+d_2-2)$ for $x_1^{d_1-1}y_1x_2^{d_2-1}y_2$ with $d_1,d_2 \ge 2$ in Theorem 1.1. Thus the two-block specialization is already present in that work; its section on three or more factors leaves that case open.

When all $a_i = b_i$, the lists $\mathscr{V}_i$ are absent, the formula reduces to $R_{\mathbf{d}}\big(\prod_i x_i^{b_i}y_i^{b_i}\big) = \prod_i(b_i+1)$, and the lattice matrix Δ is diagonal.

Example 8.1 (Ranks 20 and 54)**.** For the cubic triple $x_1^2y_1x_2^2y_2x_3^2y_3$ of Example 1.1, $b_i = 1$ and $\delta_i = 1$, so $N = 2^3 + 3 \cdot 2^2 = 20$. The matrix Δ has diagonal entries 3 and off-diagonal entries 1, and (58) is a sum over the 20 characters of $\mathbb{Z}^3/\Delta\mathbb{Z}^3$. For $\mathbf{a} = (3,3,3)$ and $\mathbf{b} = (2,2,2)$, we have $\delta_i = 1$, so

$$R_{(5,5,5)}\big(x_1^3y_1^2x_2^3y_2^2x_3^3y_3^2\big) = 3^3 + 3 \cdot 3^2 = 54.$$

Here Δ has diagonal entries 4 and off-diagonal entries 1, and (58) provides the 54 summands.

Remark 8.2 (Pure-power factors)**.** The hypothesis $b_i \geq 1$ is essential to Theorem 8.1 as stated. In the lower bound it is used where the remainder in (48) is evaluated at a zero coordinate. The formula would already be false for a single pure power x^a with $b = 0$ and $a \geq 1$: its rank is 1, whereas the right-hand side of (37) would be $a + 1$. Such factors are excluded from the statement.

A Coordinate factors in the Gram entries

This appendix records the cancellation and degree checks used in Lemma 7.1. For a basic function indexed by $\jmath$ and a function from $\mathscr{B}_i$ indexed by (s, ι), their product with H_t is

$$P_{ab}(Z) = Z_i^{s+\jmath_i} h_i(Z_i)^{d_i-s-1} \prod_{j\neq i} Z_j^{\iota_j+\jmath_j} h_j(Z_j)^{d_j-2}. \tag{60}$$

The exponent $d_i - s - 1$ is nonnegative because $s \leq d_i - 1$. The degrees in coordinate i range from $s + \jmath_i \geq 2$ to $d_i - 1 + \jmath_i \leq d_i$. Thus the ith factor has zero L_i-value. Each other factor has degree at most d_j. Applying (12) gives $L(P_{ab}) = 0$.

Next choose functions from $\mathscr{B}_i$ and $\mathscr{B}_j$, with $i \neq j$. Let their indices be (s, ι) and $(s', \jmath)$, respectively. Then

$$\begin{aligned} P_{ab}(Z) = {} & Z_i^{s+\jmath_i} h_i(Z_i)^{d_i-s-1} Z_j^{s'+\iota_j} h_j(Z_j)^{d_j-s'-1} \\ & \times \prod_{m\neq i,j} Z_m^{\iota_m+\jmath_m} h_m(Z_m)^{d_m-2}. \end{aligned} \tag{61}$$

Both distinguished coordinate factors have nonnegative powers of h_i, h_j and monomial degrees between 2 and their respective degree bounds. All remaining factors are also within their bounds. The product rule therefore gives $L(P_{ab}) = 0$.

Finally, for two functions from the same family $\mathscr{B}_i$, the coordinate factors are

$$P_{ab}(Z) = Z_i^{s+s'} h_i(Z_i)^{d_i-s-s'} \prod_{j\neq i} Z_j^{\iota_j+\jmath_j} h_j(Z_j)^{d_j-2}. \tag{62}$$

This is polynomial when $s + s' \leq d_i$, yielding the zero blocks in Lemma 7.2. At $s + s' = d_i + 1$, one denominator remains, yielding H_i in (29). In particular, this last case is not an application of the polynomial moment formula to a rational function; it requires the calculation below.

B The leading coefficient of the rational blocks

We give the exact identity underlying Lemma 7.3. This also shows explicitly why zero coordinates of the decomposition points cause no difficulty.

Lemma B.1 (Finite division identity)**.** *For an integer $d \geq 1$, $t \neq 0$, and $1 + tz \neq 0$,*

$$\frac{z^{d+1}}{1+tz} = \sum_{n=0}^{d-1} \frac{(-1)^n z^{d-n}}{t^{n+1}} + \frac{(-1)^d z}{t^d(1+tz)}. \tag{63}$$

For each fixed $z \in \mathbb{C}$, the final term is $O(t^{-d-1})$ as $t \to \infty$; when $z = 0$, every term is identically zero.

Proof. Multiply by $1 + tz$. The consecutive terms in the finite sum cancel, leaving z^{d+1} after the remainder is included. For $z \neq 0$, $z/(1 + tz) = O(t^{-1})$; for $z = 0$ it is zero. □

Detailed proof of Lemma 7.3. Fix i and treat t_i as the variable, keeping the other parameters fixed. For row and column indices $\iota, \jmath$, set $A_{\iota,\jmath}(Z_{-i}) = \prod_{j\neq i} Z_j^{\iota_j+\jmath_j}(1 + t_jZ_j)^{d_j-2}$. Each coordinate degree of $A_{\iota,\jmath}$ is at most d_j. Substituting (63) with $d = d_i$ into (29) gives

$$[H_i(t)]_{\iota,\jmath} = \sum_{n=0}^{d_i-1} (-1)^n t_i^{-n-1} L\Big(Z_i^{d_i-n} A_{\iota,\jmath}(Z_{-i})\Big) + O(t_i^{-d_i-1}). \tag{64}$$

The remainder estimate is valid because the sum over the fixed points is finite. The finite identity applies also to points with $z_{i\ell} = 0$; no division by such a coordinate is used.

For $n = 0, \dots, d_i - 2$, the ith exponent $d_i - n$ lies between 2 and d_i. The corresponding L-value is zero by (12). For $n = d_i - 1$, the same rule gives

$$L(Z_i A_{\iota,\jmath}(Z_{-i})) = \beta_i \prod_{j \neq i} L_j\left(Z_j^{\iota_j+\jmath_j}(1+t_j Z_j)^{d_j-2}\right) = \beta_i \left[\bigotimes_{j\neq i} G_j(t_j)\right]_{\iota,\jmath}. \tag{65}$$

Equations (64) and (65) prove (33).

For completeness, put $C_i = (-1)^{d_i-1}\beta_i \bigotimes_{j\neq i} G_j(t_j)$. This $q \times q$ matrix is invertible. Taking determinants gives

$$\det H_i(t) = t_i^{-d_i q}\left(\det C_i + O(t_i^{-1})\right), \qquad q = 2^{k-1}, \quad \det C_i \neq 0. \tag{66}$$

Thus $\det H_i$ cannot be the zero rational function: the entries of H_i have denominators only of the form $1+t_i z_{i\ell}$, so the other t_j may be specialized to arbitrary complex values, and a nonzero specialization gives $\det H_i \neq 0$ in $\mathbb{C}(t)$. Equivalently, the calculation can be carried out as a formal Laurent expansion over the field $\mathbb{C}(t_j : j \neq i)$. □

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Data availability. The exact verification scripts and Lean 4 files accompanying this paper are available at https://github.com/erenup/w-tensor-product-rank.

Use of generative AI and AI-assisted technologies. During the preparation of this work, the authors used AI tools (Codex and Claude Code) to generate candidate constructions and code, perform computations, draft mathematical arguments, and improve the readability of the text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The lower-bound argument for Theorem 1.1 is a complete written proof; finite instances of its coefficient identities were additionally checked in exact arithmetic, and a Lean 4 formalization covers six of its algebraic ingredients and its one-coordinate case under explicit hypotheses, as stated in the repository above, while the test functions, the cross-block cancellations, the identification of each block H_i with its formal expansion and the reduction to moments are proved in the text. Section 8 is a complete written proof, with exact checks of finite instances.

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